

Inverse Problems in Geophysics

Part 13: Summary non-linear

2. MGPY+MGIN

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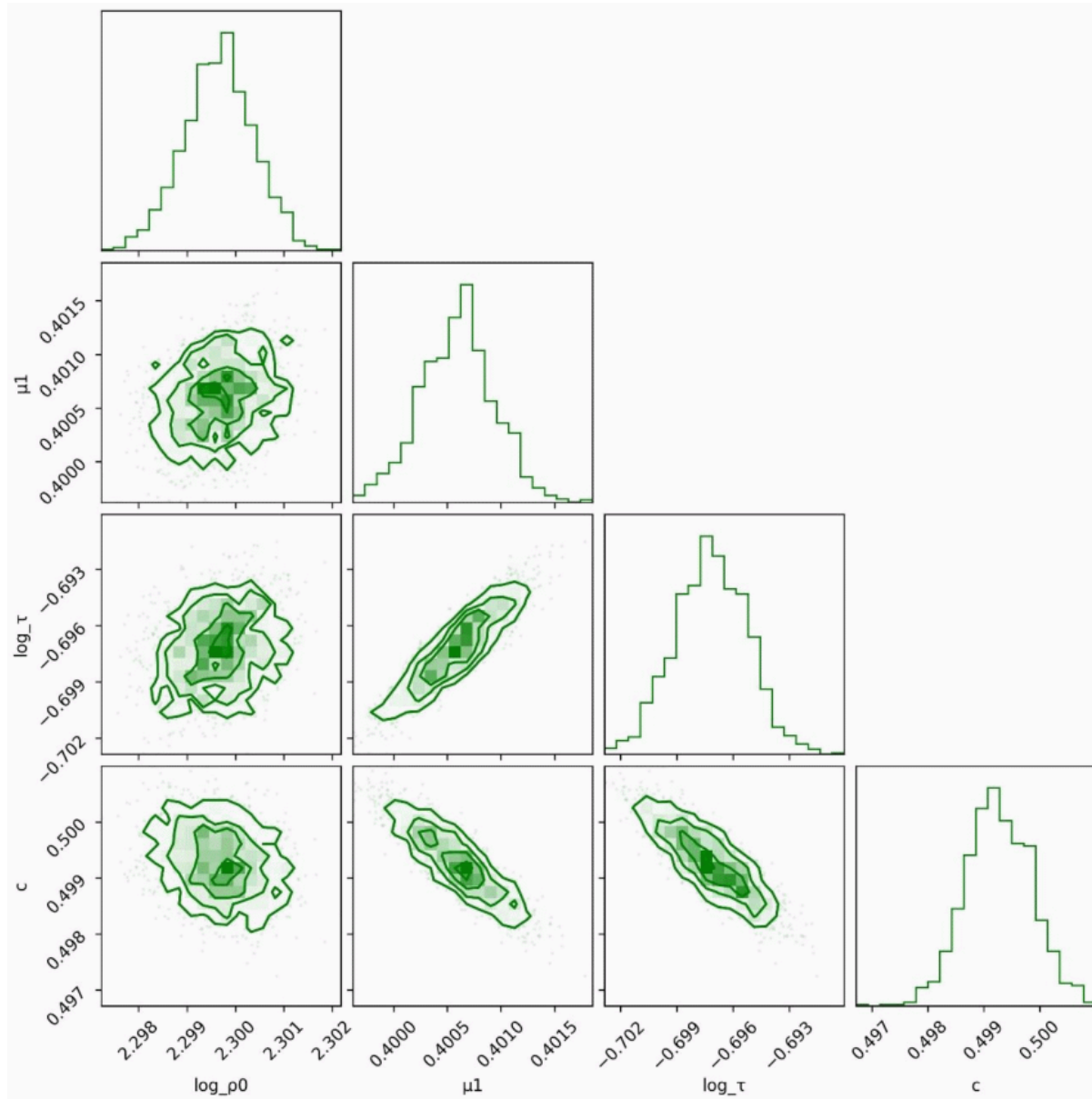
Previously: Excursion into global minimization

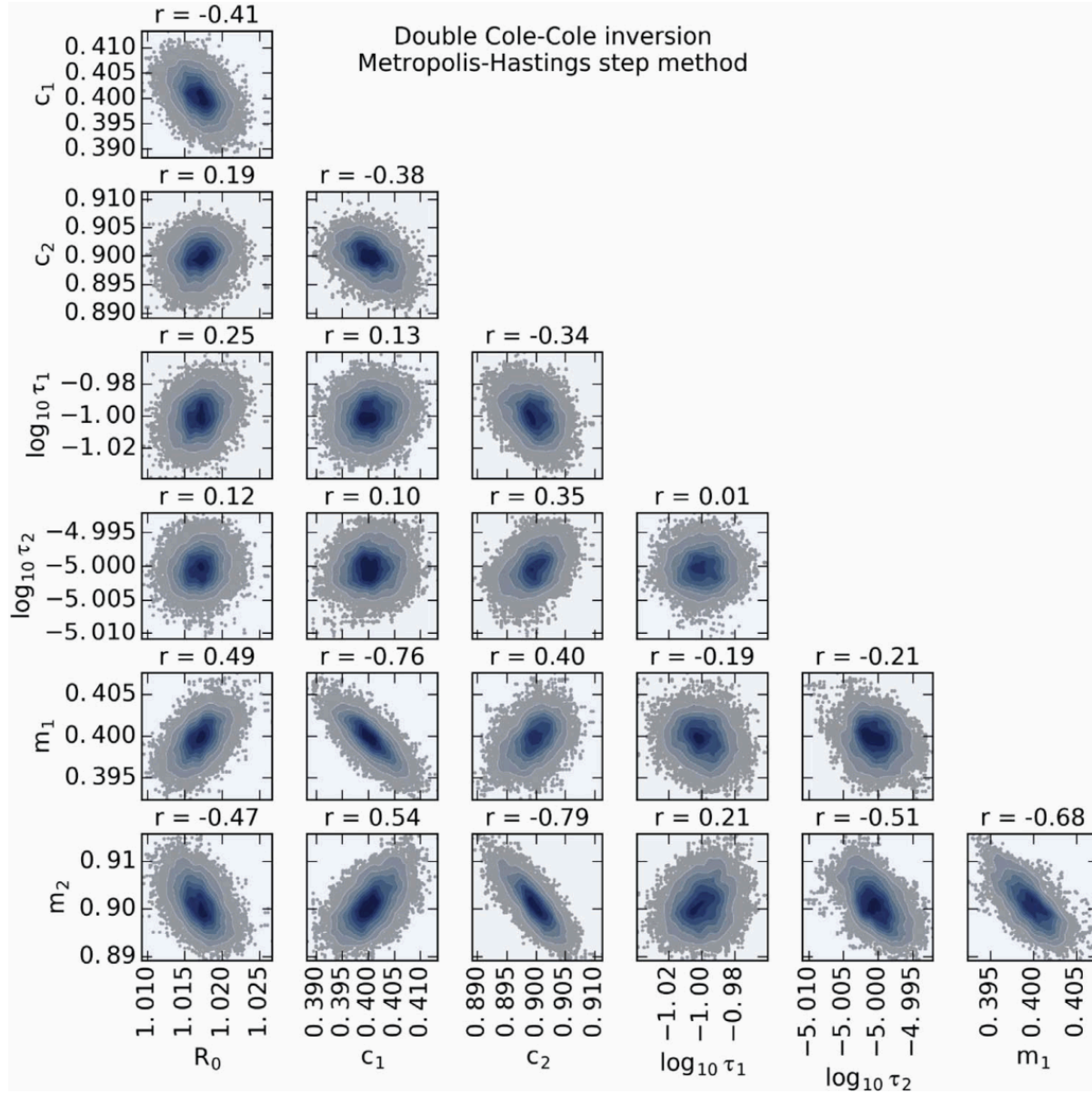
- Bayes is connected to regularized inversion:
 - prior distribution is regularization
 - likelihood describes forward problem and errors
 - posterior solution is trade-off between both
- Monte-Carlo family of global minimization
- Metropolis-Hastings algorithm
- Markov-chain Monte Carlo (MCMC) delivers groups of models
 - trace plots show convergence, corner plots show correlation

Corner plots

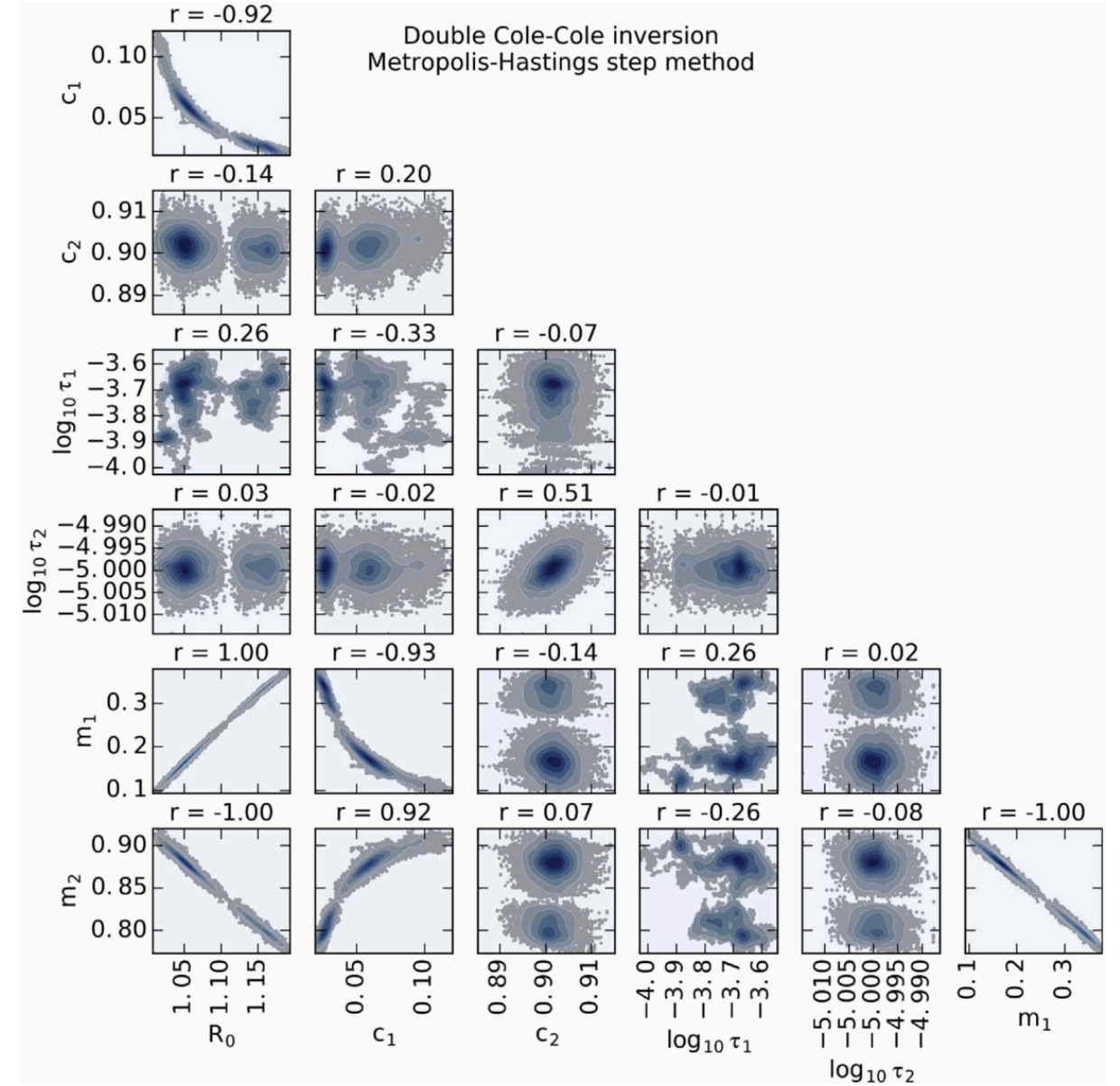
(Roudsari et al., in rev.)

- statistics of models
- correlation of parameters
- uncertainty of the results



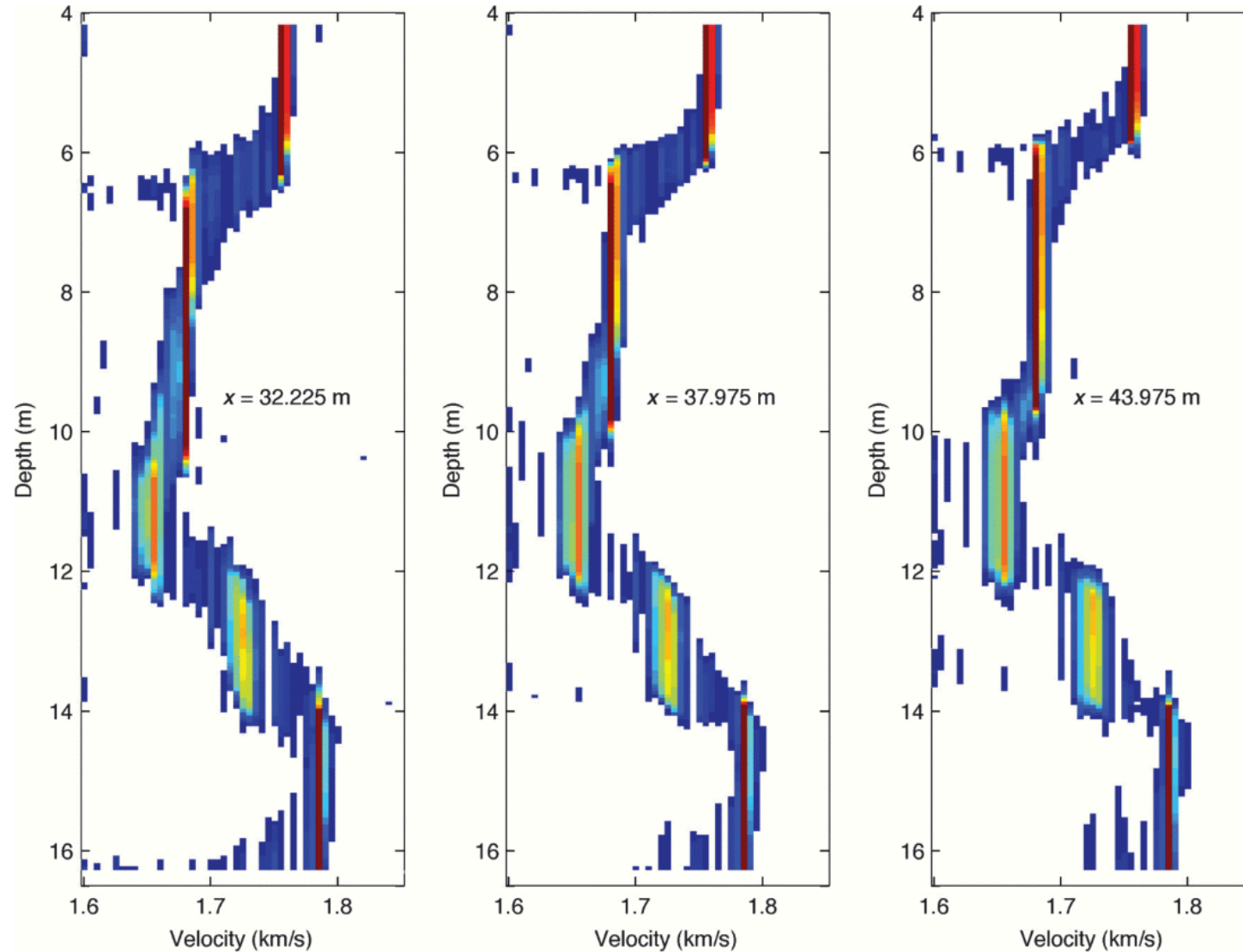


well-defined (Berube et al. 2017)



poorly defined (Berube et al. 2017)

Model variance



- derivation of mean values and standard deviations
- statistic simulations

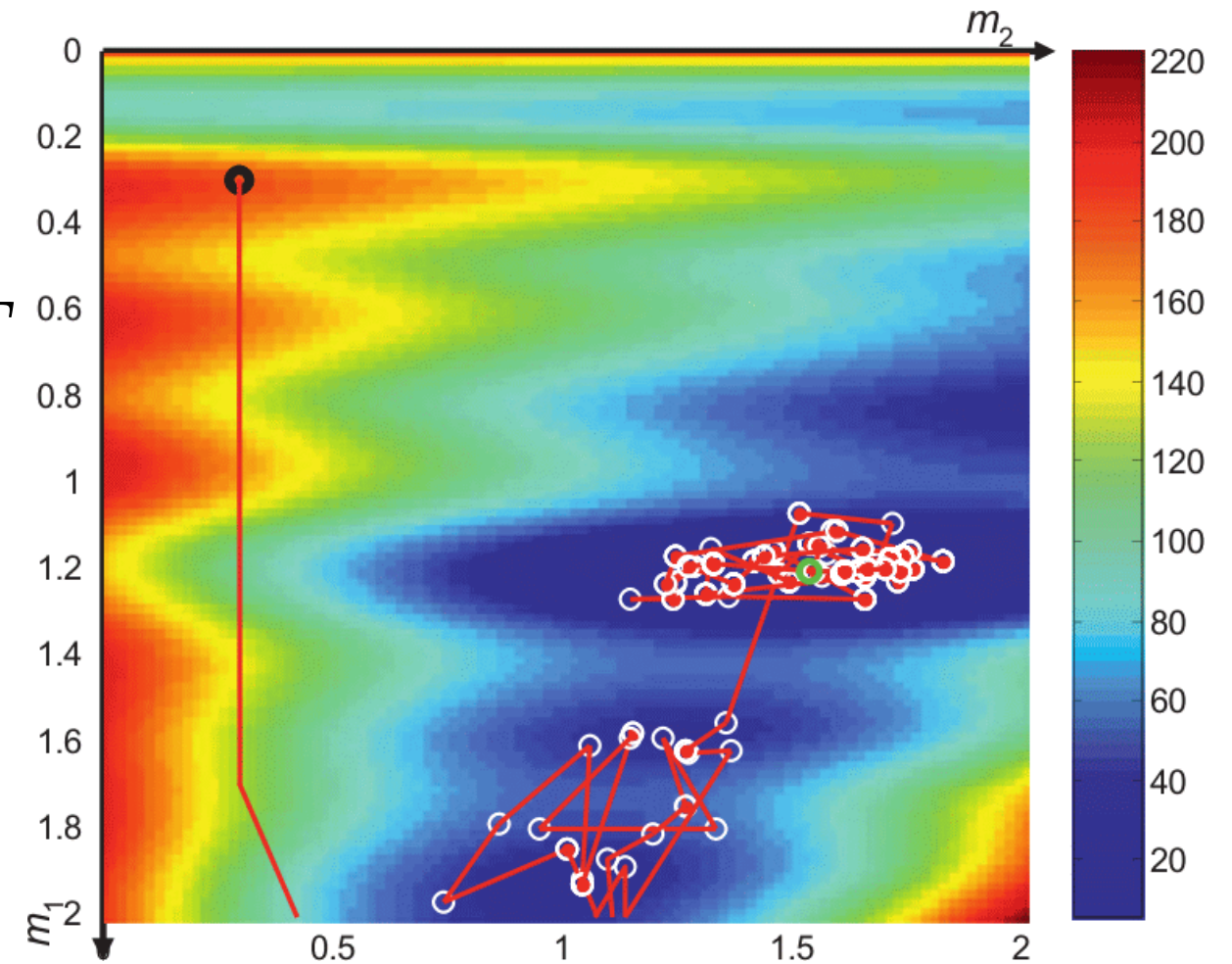
(Tronicke et al., 2012)

Simulated Annealing

Gibbs (Boltzmann) sampling

$$P(E) = e^{-\frac{E}{k_B T}} = e^{-(\Phi(\mathbf{m}) - \Phi(\mathbf{m}^p))/T}$$

- draw models neighboring best fit
- large movements if far from target fit
- small if approaching target

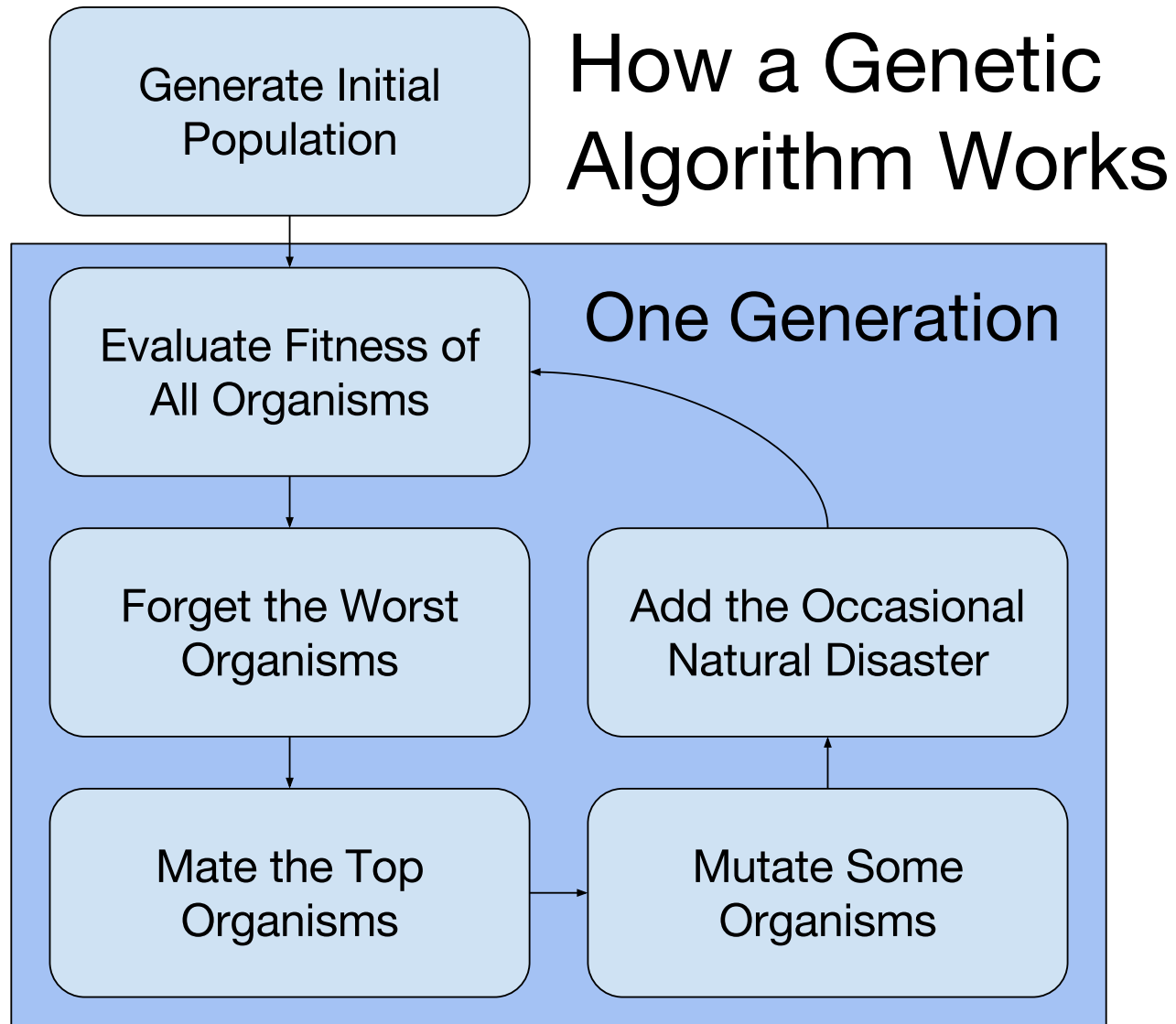


Simulated Annealing

Bio-inspired global search methods

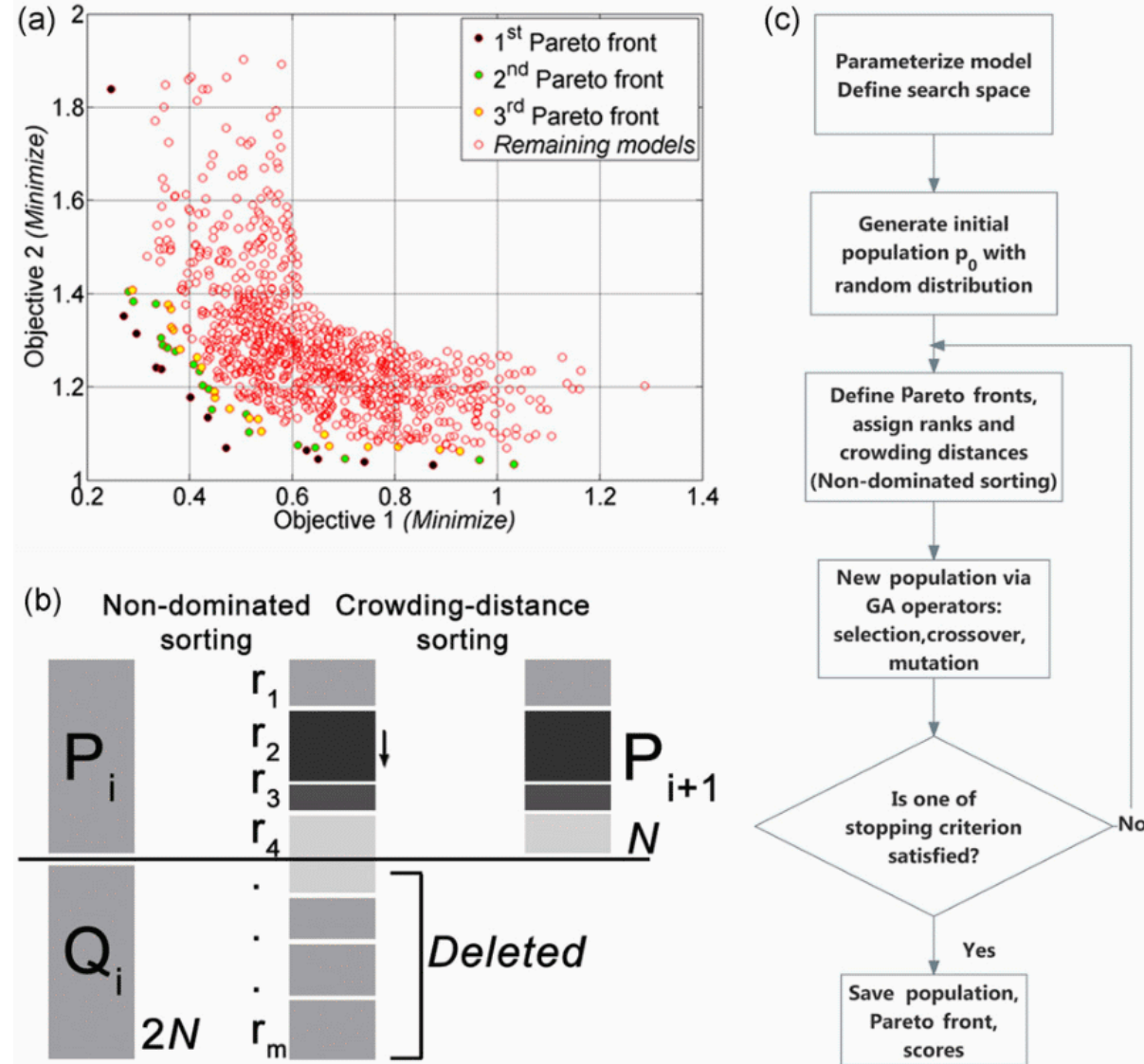
- Genetic Algorithm
- Evolution Strategy
- Differential Evolution Algorithm
- Estimation of Distribution Algorithm
- Particle Swarm Optimization
- Ant Colony Optimization
- Pareto Archived Evolution Strategy (PAES)
- Nondominated Sorting Genetic Algorithm (NSGA-II)

Genetic algorithms



- inspired by evolution of species (*survival of the fittest*)
- code models in a binary (DNA) sequence
- randomly generate initial population
- let the fittest (data fit!) survive & mate and the unfittest die
- mutation of the DNA to create variability

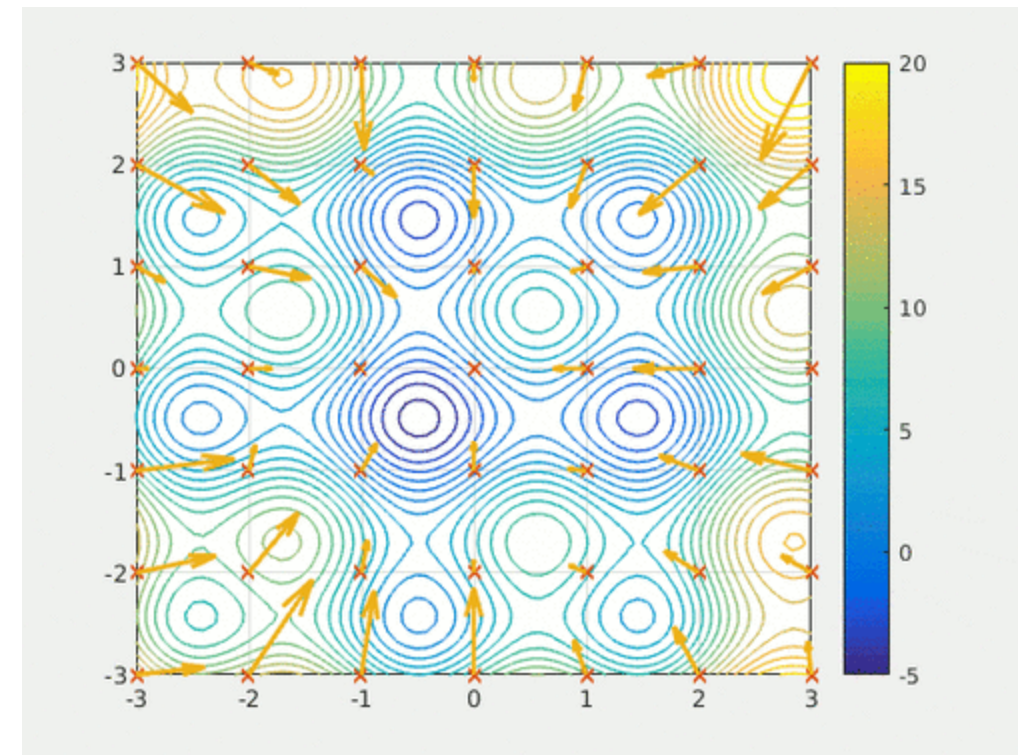
Nondominated Sorting Genetic Algorithm



- minimization of two objective functions
 - two geophysical methods
 - data misfit and model roughness
- Pareto front: all non-dominated individuals
- sort Pareto rank and play god

Particle swarm optimization

- movements of individuals in swarm (impulse & attraction)
- individual position ($\mathbf{m}$) & velocity
- distribute population over range
- acceleration from attraction
- move with velocity



Summary

global (probabilistic) inversion methods

- can overcome local minimum problems
- also find a trade-off between data fit and model assumptions
- find a variety of models fitting the data
- can therefore show correlation and model uncertainty
- only applicable to “small” problems (increasingly simple 2D)
- well suited as supplement to local methods

Back to deterministic inversion

- iterative solution beginning from starting model
-

Levenberg-Marquardt algorithm

local damping of $\Delta \mathbf{m}$ (for stabilizing only)

$$\Delta \mathbf{m} = (\mathbf{J}^T \mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{J}^T (\mathbf{d} - \mathbf{f}(\mathbf{m}))$$

typically used for independent parameters:

- coefficients of fitting (e.g. petrophysical) models
- physical parameters & layer thicknesses of (few) layered model
- often used with logarithmic parameters preventing negatives
- λ usually decreased, i.e. $\lambda^k = \lambda^0 v^k$ (e.g., $v=0.5$ etc.)

Occam (smoothness-constrained) inversion

global regularization of the smoothness $\|\mathbf{C}\mathbf{m}\|$

$$\Delta\mathbf{m} = (\mathbf{J}^T\mathbf{J} + \lambda\mathbf{C}^T\mathbf{C})^{-1}(\mathbf{J}^T(\mathbf{d} - \mathbf{f}(\mathbf{m})) - \lambda\mathbf{C}^T\mathbf{C}\mathbf{m}))$$

used for pre-discretized models: 1D, 2D, 3D, 4D

- full application of Gauss-Newton minimization
- different types of smoothness: 1st/2nd order, geostatistics
- decoupling different regions, region-wise regularization etc.
- regularization parameter λ often kept constant or slightly cooled

Resolution matrices for non-linear problems

$$\Rightarrow \mathbf{m}^{est} = \mathbf{m}^k + \mathbf{J}^\dagger \mathbf{J} \mathbf{m}^{true} - (\mathbf{J}^\dagger \mathbf{J} + \mathbf{C}^\dagger \mathbf{C}) \mathbf{m}^k + \mathbf{C}^\dagger \mathbf{C} \mathbf{m}^0 + \mathbf{J}^\dagger \mathbf{n}$$

$$\Rightarrow \mathbf{m}^{est} = \mathbf{R}^M \mathbf{m}^{true} + (\mathbf{I} - \mathbf{R}^M) \mathbf{m}^0 + \mathbf{J}^\dagger \mathbf{n}$$

$$\mathbf{R}^M = \mathbf{J}^\dagger \mathbf{J}$$

$$\mathbf{f}(\mathbf{m}^{k+1}) = \mathbf{J}(\mathbf{J}^\dagger(\mathbf{f}(\mathbf{m}^{true}) + \mathbf{d} - \mathbf{f}(\mathbf{m}^k))) + \mathbf{C}^\dagger \mathbf{C}(\mathbf{m}^0 - \mathbf{m}^k)$$

$$\mathbf{R}^D = \mathbf{J} \mathbf{J}^\dagger$$

Model covariance: $MC\mathbf{M} = (\mathbf{J}^\top \mathbf{J})^{-1}$

Example Vertical Electric Sounding

https://tubaf-em.github.io/Inversion_Geophysics/13_exampleVES.html

- Levenberg-Marquardt inversion for 3-layer case
- smoothness-constrained multi-layer inversion
- basis for report (TEM instead of VES)